ABHILASH ANTONY

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PROFESSIONAL OVERVIEW

Associate Software Engineer with a masters in Data Analytics and hands-on experience building **Generative AI, agentic AI, and RAG-based applications** using Python, FastAPI, LangGraph, vector databases, Google Gemini, and AWS. Experienced in developing AI-powered automation systems, integrating LLMs, MCPs and APIs, and designing cloud-based solutions across social media automation, knowledge retrieval, and computer vision use cases. Strong in problem-solving, technical research, solution architecture, client communication, and translating business requirements into practical AI solutions.

EXPERIENCES

Associate Software Engineer (AI and Data Science) at **Techversant Infotech Pvt. Ltd.** (link) 03 Nov 2025 – Present

- Built an **end-to-end AI-driven Instagram automation solution for a client** using **Meta Graph API, webhooks, AWS Lambda, SQS, RDS, and LLMs**, supporting automated comment replies, DMs, private replies, and direct/scheduled post publishing; subsequently **optimized the architecture by consolidating 7 Lambda functions and 6 SQS queues into a unified 1-Lambda/1-SQS design**, achieving benchmarked **~58.6% lower comment-flow latency** and **~60% lower DM-flow latency** while reducing infrastructure complexity and cost.
- Built a **standalone AI-powered social media automation and content generation platform** using **FastAPI, Next.js, PostgreSQL, AWS, Google Gemini, and Meta Graph API**, integrating campaign management, platform-specific content generation, reusable media assets, AI image generation, and end-to-end Instagram automation for **comments, DMs, private replies, and direct/scheduled post publishing**.
- **Worked with the team to build an agentic RAG chatbot** using **Qdrant, LangGraph, LangSmith, and Langfuse**, contributing to multi-agent workflows and tool integrations for **property booking, concierge assistance, and contextual AI responses**, with tracing and observability across agent execution.
- **Worked on a client project** contributing to the migration of its API-based data processing architecture to an **ETL pipeline using AWS Glue and PySpark**, including **RDS integration, data consolidation, transformations, and SQS-based data synchronization**.
- Applied software-engineering best practices across development, including **Git/GitHub workflows, code reviews, modular architecture, API design, validation, error handling, testing, and maintainable code practices**.
- **Collaborated directly with clients, stakeholders, and cross-functional teams** to understand requirements, evaluate AI/technical feasibility, discuss API and platform limitations, present solution architectures and demos, and prepare **BRDs, feasibility studies, technical documentation, architecture diagrams, and implementation plans** to guide project execution.

Data Engineer Intern at **SKYNETCLOUDS CO. LTD, Bangkok** (link) 11 Dec 2024 – 20 March 2025

- Developed an **AI-driven breast cancer detection system** from mammogram files using **Python, TensorFlow, CNNs, and ResNet101 transfer learning**, building an end-to-end pipeline covering data preparation, augmentation, model training, optimization, and evaluation.
- Evaluated multiple **deep-learning, classical ML, anomaly-detection, and image-segmentation approaches** before selecting and optimizing ResNet101; addressed **class imbalance and false negatives** using data augmentation, class weighting, threshold optimization, dropout, early stopping, and multi-GPU training, achieving a reported **98.76% test accuracy**.
- Integrated the trained model into a **Streamlit proof-of-concept application** supporting mammogram upload, preprocessing, real-time cancer/non-cancer prediction, confidence scoring, and downloadable diagnostic reports.

Data Science Intern at **Evoastra Ventures Pvt. Ltd., Bengaluru** (link) 25 July 2024 – 25 August 2024

- **Worked on a used-car market analysis project** by contributing to automated **Python and Beautiful Soup web scraping** of Cars24 listings, followed by data cleaning, exploratory analysis, and visualization to identify **pricing, model popularity, fuel-type, transmission, mileage, and year-wise trends** across Tata, Hyundai, and Maruti.
- **Contributed to a credit-risk analysis project** by performing **univariate, bivariate, and correlation-based EDA** on historical loan applications to identify **approval trends, cancellation drivers, customer behavior patterns, and higher-risk applicant profiles**, translating findings into recommendations for improving loan approval, reducing cancellations, and mitigating financial risk.

PROJECTS

PHISHING WEBSITE DETECTION USING ML (link) July 2024 – August 2024
TechStack: Python, Scikit-learn, Pandas, NumPy, Matplotlib
Description: Built and optimized ML models for phishing detection using website URL features, achieving **97% accuracy** with a tuned Random Forest model. Conducted feature analysis (SSL, URL length, IP presence, etc.), handled class imbalance, and improved reliability through hyperparameter tuning with GridSearchCV.

MUTUAL FUND PORTFOLIO OPTIMISATON (link) August 2024 – September 2024
TechStack: Python, Pandas, NumPy, Plotly, Jupyter Notebook
Description: Analyzed Nifty50 stock price trends to assess volatility, simulate SIP investments, and evaluate long-term returns. Built a diversified mutual fund portfolio, visualized growth patterns, and demonstrated the benefits of risk-aware, compounding-based investment strategies.

NETFLIX CONTENT STRATEGY ANALYSIS (link) October 2024 – November 2024
TechStack: Python, Pandas, Matplotlib, Seaborn
Description: Analyzed Netflix's 2023 content trends to evaluate the impact of content type, language, release timing, and holidays on viewership. Discovered that shows dominated engagement, English and Korean titles led globally, and Friday/holiday releases maximized audience retention.

CASHFLOW – PERSONAL EXPENSE TRACKER July 2025 – August 2025
TechStack: Django, Python, SQLite, HTML, CSS, Bootstrap
Description: Developed a full-stack web application to manage personal finances. Implemented features for adding, editing, deleting, and restoring expense entries, with category and type-based filters. Integrated date-range filtering, transaction statements, and custom confirmation prompts for secure deletions. Designed a clean, responsive UI with Bootstrap, gradient branding, and a custom logo.

TECHNICAL SKILLS

Languages: Python, R, SQL, HTML, CSS
Technologies/Frameworks/Libraries: Power BI, MS Excel, MS Word, Google Sheets, Google Docs, Hadoop, PIG, Streamlit, GitHub, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, NLTK, RDBMS, MySQL, MongoDB, Django.
Course Works: Data Analysis, Machine Learning, Deep Learning, NLP, Generative AI, Data Visualization, Database Management, Statistics, Mathematical Modelling, Data Structures & Algorithms, R Programming, Big Data, Statistics & Probability, Advanced Excel, Data Visualization, Data Cleaning, Linear Algebra, Web Development.
Others: Team Work, Communication, Problem Solving, Critical Thinking, Leadership, Teamwork, Communication, Research & Reporting, Problem-Solving.

EDUCATION

CHRIST (DEEMED TO BE UNIVERSITY), BANGALORE, KARNATAKA (link) July 2023 – May 2025
MSc. Data Analytics CGPA: 9.4/10.0

MARY MATHA ARTS & SCIENCE COLLEGE, WAYANAD, KERALA (link) June 2020 – May 2023
BSc. Mathematics, Statistics, Computer Science CGPA: 8.0/10.0

CERTIFICATIONS

GOOGLE DATA ANALYTICS PROFESSIONAL CERTIFICATE (link) October 2023 – March 2024
Gained hands-on training in data cleaning, analysis, and visualization using SQL, R, Spreadsheets, and Tableau, with practical experience in organizing, analyzing, and presenting data insights.